



KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN
MECHANICAL ENGINEERING
EMM 205: FLUID MECHANICS II

1ST SEMESTER 2022/2023

TIME: 2 HOURS

INSTRUCTIONS:

- i. This paper contains FIVE Questions.
- ii. Question ONE is compulsory and carries 30 marks.
- iii. Attempt Question ONE and any other TWO Questions from the remaining FOUR Questions. FOUR remaining Questions carry 20 marks each.

Question One (Compulsory 30 marks)

- (a) (i) What is the physical meaning of Reynolds's Number? (2 marks)
(ii) Why is it critical in flow fluids? (2 marks)
- (b) A horizontal jet of cross sectional area am^2 , mass density ρ_{-3m}^{kg} moving at a velocity of Vm/sec strikes an inclined plate at angle of θ degrees to the normal to the plate. Derive an expression for the normal force exerted on the plate when it is stationary. (2 marks)
- (c) If the plate is moving at um/sec in the direction normal to the plate derive an expression for the
 - (i) force exerted on the plate (2 marks)
 - (ii) work done per second (2 marks)
 - (iii) efficiency of the arrangement (2 marks)
- (a) A flat plate is struck at an angle of 60° to the normal to the plate by a jet of water 50mm in diameter with a velocity of 18m/sec. Calculate the:
 - (i) Normal force exerted on the plate when it is stationary (2 marks)
 - (ii) Normal force exerted on the plate when it is moving to the direction normal to the plate at a velocity of 6m/sec. (4 marks)
 - (iii) work done per second (4 marks)
 - (iv) Efficiency of the arrangement. (5 marks)

QUESTION TWO (20 MARKS)

- (a) (i) Working from first principles and explaining the meaning of all symbols used, derive an expression for the rate of steady flow under lamina conditions of a viscous incompressible fluid through a rectangular passage of width b , a small depth h and a length L in the direction of flow due to pressure difference p .
- (ii) Show that shear stress on the wall τ is given by

$$\tau = \frac{6\eta V}{h} \quad \text{where } \eta = \text{coefficient of dynamic viscosity}$$

$V = \text{mean velocity of flow}$

- (b) The radial clearance between a plunger and the walls of a cylinder is 0.075mm, the length of the plunger is 250mm and its diameter 100mm. The pressure difference between the ends of the plunger is 207kN/m².

Estimate the rate of leakage in litres per sec.

Assume $\eta_{\text{water}} = 1.31 \times 10^{-3} \frac{\text{kg}}{\text{m sec}}$ (6 marks)

QUESTION THREE (20 MARKS)

- (a) Show that the loss of head when a pipe undergoes a sudden increase in diameter (H_L) is given by

$$H_L = \frac{(V_1 - V_2)^2}{2g} \quad \text{Where } V_1 = \text{velocity in the small pipe upstream of the enlargement.}$$

$V_2 = \text{velocity in the larger pipe}$
 $g = \text{acceleration due to gravity}$

(8 marks)

- (b) A pipe 344mm in diameter carries a discharge of 283 dm³/sec of water. The pipe enlarges suddenly to 688mm in diameter. The pressure in the smaller pipe is 4.8kN/m²

Calculate:

- (i) Head lost due to enlargement (4 marks)
- (ii) Pressure at the larger portion. (4 marks)
- (iii) Power needed to overcome the loss (4 marks)

QUESTION FOUR (20 MARKS)

Water is discharged to the atmosphere through a pipe 40m long. The first 15m of the pipe is 50mm in diameter and then suddenly enlarges to 75mm for the remaining length 25m.

Taking into account the losses at entry, due to friction, sudden enlargement and exit, determine head causing flow if the discharge is $0.0028 \text{ m}^3/\text{sec}$. (4 marks)

Assume: $f = 0.005$ for the 50mm diameter pipe

$f = 0.006$ for the 75mm diameter pipe (20 marks)

QUESTION FIVE (20 MARKS)

(a) What do you understand by the terms:

(i) Dynamic similarity (2 marks)

(ii) Corresponding speed (2 marks)

(b) Show that the wave making resistance (R) of a ship's hull may be expressed

$$R = \ell v^2 L^2 f \left(\frac{v}{\sqrt{gl}} \right) \quad \text{Where } \ell = \text{mass density of water.}$$

v = velocity in m/sec

L = characteristic length

g = acceleration due to gravity

(c) A ship is to be designed to cruise at 12.5 m/sec . At the corresponding speed" the wave

making component of resistance of a model to $\frac{1}{40}$ scale is far to be 16 N when tested in fresh water.

Calculate the

(i) Model speed (2 marks)

(ii) Wave making resistant of the prototype in sea water with mass density

1025 kg/m^3 .

(20 marks)



KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE (MECHANICAL ENGINEERING)

EMM 211: ENGINEERING MECHANICS II

1st SEMESTER 2022/2023

TIME: 2 HOURS

INSTRUCTIONS:

- i. This paper consists of FIVE questions
- ii. Question ONE is compulsory and contains 30 marks
- iii. Attempt any TWO of the remaining four questions. Each question is 20 marks
- iv. All symbols have their usual meaning unless otherwise specified
- v. Take acceleration due to gravity $g = 9.81 \text{ m/s}^2$

Question One (30 marks)

- (a) A ship in water is being steered to change its direction of motion. During the process of steering, the ship can be assumed to follow a circular path of radius 100m. The velocity of the ship is being reduced uniformly and is reduced from 20 km/hr to 15 km/hr within 50 seconds. Determine the total acceleration of the ship at the two points. [5 Marks]
- (b) During a test a rocket is travelling upwards at 75m/s, and when it is 40m from the ground its engine fails. Determine the maximum height reached by the rocket and its speed just before it hits the ground. While in motion, the rocket is subjected to a constant downward acceleration of 9.81 m/s^2 due to gravity. Neglect the effect of air resistance. [5 Marks]
- (c) Two cars move from a port at the same time. Car A has a velocity of 30km/hr and is moving $W 60^\circ N$ while car B moves in south-west direction with a velocity of 40km/hr. Determine; [8 Marks]
- i. The relative velocity of A with respect to B. [2 Marks]
 - ii. The distance between the ships after 10 minutes.
- (d) Distinguish between the principle of linear impulse and momentum and angular impulse and momentum, and state relevant applications of the same. [4 Marks]
- (e) A disk is driven by a motor such that the angular position of the disk is defined by $\theta = (20t + 4t^2)$ rad, while t is time in seconds. Determine the number of revolutions, the angular velocity, and the angular acceleration of the disk after 90 seconds. [6 Marks]

Question Two (20 marks)

- (a) Define the term radius of gyration as used in Engineering Mechanics. [2 Marks]
- (b) Show that the moment of inertia of the cylinder shown in Figure 1 below about the z axis if the density of the material (ρ) is constant is given as $I_z = \frac{1}{2}mR^2$ where m and R are the total mass and the radius of the cylinder respectively. [7 Marks]

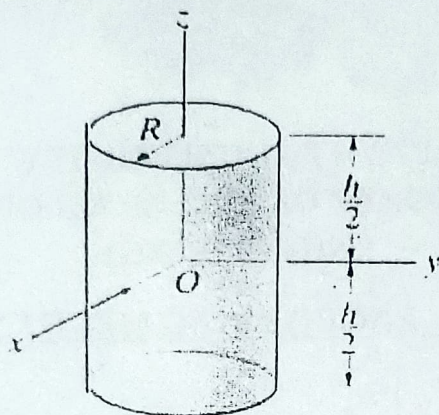


Figure 1

- (c) A motor as shown in Figure 2 supplies a constant torque of 2 Nm to a 50mm diameter shaft O connected to the center of a 30kg flywheel. The resultant bearing friction F, which the bearing exerts on the shaft, acts tangent to the shaft and has a magnitude of 50N. Determine how long the torque must be applied to the shaft to increase the flywheel's angular velocity from 4rad/s to 5rad/s. The flywheel has a radius of gyration $k_o = 0.15m$ about its center O. [6 Marks]

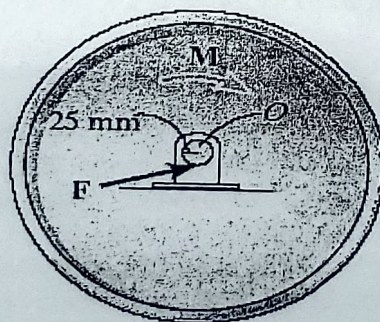


Figure 2

- (d) A pulley of weight 400N has a radius of 0.6m. A block of 600N is suspended by a tight rope wound round the pulley. Determine the resulting acceleration of the block and tension in the rope. Assume that the moment of inertia of the pulley about the center of rotation can be estimated by $\frac{mR^2}{2}$ where R and m are the radius and mass of the pulley respectively. [5 Marks]

Question Three (20 marks)

- (a) Define the following quantities and give their expressions [3 Marks]
- Potential Energy
 - Kinetic Energy
 - Strain Energy
- (b) State the principle of linear impulse and momentum, and the principle of work and energy, as applied in kinetics. [4 Marks]

- (c) Block D shown in Figure 3 moves with a speed of m/s . Using the instantaneous center of rotation method, determine the angular velocities of link BD and AB, at the instant shown. [13 Marks]

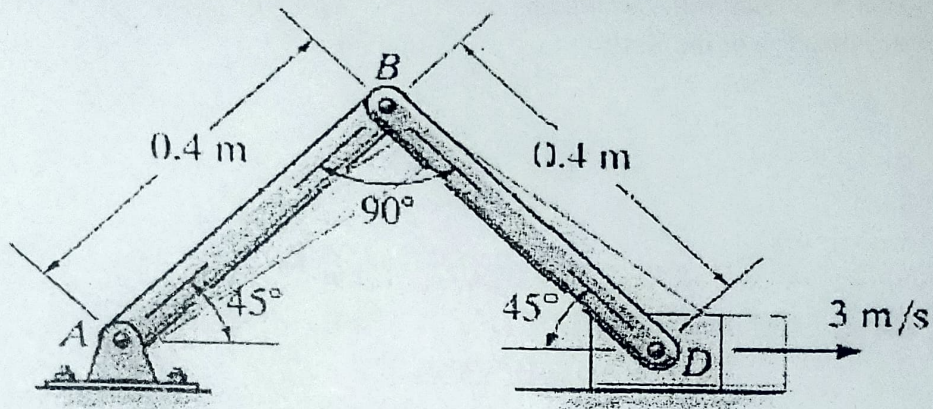


Figure 3

Question Four (20 marks)

The unloading mechanism used on a small waste disposal vehicle is shown in Figure 4 below. The length of the links $AB = 0.9m$, $BC = 0.2m$ and $CD = 1.25m$. For the instant shown, the link BC is horizontal and link CD is vertical. The link AB is rotating anti-clockwise at an angular velocity of $0.5 rad/s$ and with an anti-clockwise angular acceleration of $2 rad/s^2$.

- Using the relative velocity analysis by vector rotation, determine the angular velocities of the skip (link CD) and of link BC. [10 Marks]
- Determine the acceleration of pin B and of the skip (link CD) [10 Marks]

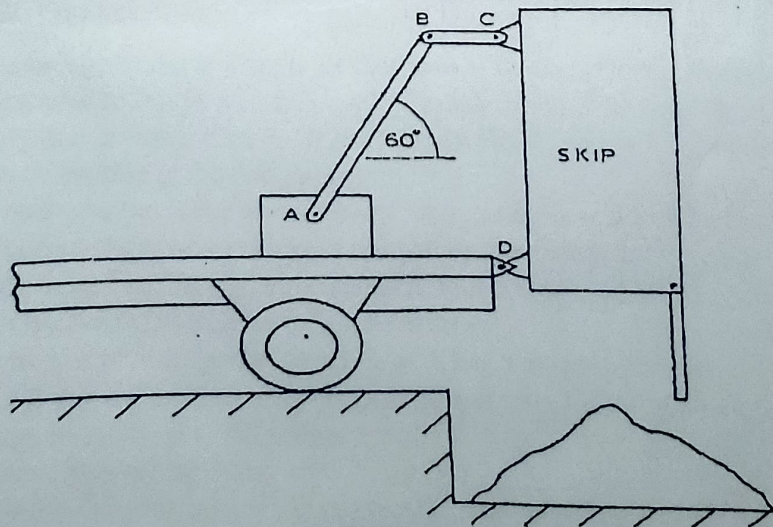


Figure 4

Question Five (20 marks)

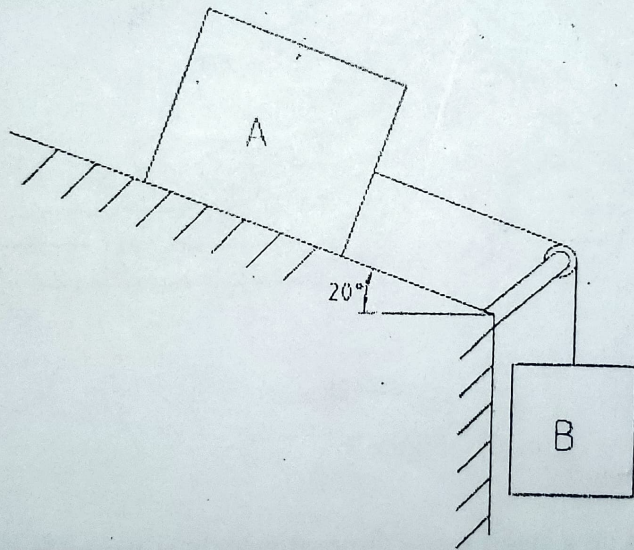
- State D'Alembert's principle as used in Engineering Mechanics. [2 Marks]
- Using relevant examples distinguish between static, limiting and dynamic friction. [3 Marks]
- A body weighing $300N$ is pushed up a 30° plane by a $400N$ force acting parallel to the plane. If the initial velocity of the body is $1.5m/s$ and the coefficient of kinetic friction is $\mu_k = 0.2$, use the principle of work and energy to determine the velocity of the body after moving $6m$. [5 Marks]

(d) Two crates shown in Figure 4 below are released from rest. Their masses are $M_A = 40\text{kg}$ and $M_B = 30\text{kg}$. If their static and kinetic coefficients of friction between crate A and the inclined surface are $\mu_s = 0.2$ and $\mu_k = 0.15$ respectively, determine;

- Whether the crates will start moving.
- The acceleration of the crates if they are in motion.

[5 Marks]

[5 Marks]





KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE (MECHANICAL ENGINEERING)

EMM 214 ELECTRICAL ENGINEERING II

1ST SEMESTER 2022/2023

TIME: 2 Hours

INSTRUCTIONS:

- i. This paper consists of **FIVE** questions
- ii. Question **ONE** is compulsory and contains 30 marks
- iii. Attempt any **TWO** of the remaining four questions. Each question is 20 marks

Question one (30 marks)

- i) What is the principle of operation of a DC Generator? (4 Marks)
- ii) A 520V, 50hp, three phase induction motor is drawing 60A at 0.95 pf lagging. The stator copper losses are 2.5kW and the rotor copper losses are 800W. The friction loss is 700W and the ~~copper~~ ^{core} losses are 1800W. Find:
 - (a) The airgap power P_{AG}
 - (b) The converted power P_{conv}
 - (c) The output power P_{OUT}
 - (c) The efficiency of the motor (4 Marks)
- iii) Differentiate between separately excited and self-excited dc generators (4 Marks)
- iv) At 200V a short-shunt compound generator supplies a current of 120 A. If the armature resistance, $R_a = 0.08 \Omega$, field resistance, $R_f = 40 \Omega$, and the series resistance, $R_{se} = 0.04 \Omega$. Determine the e.m.f. generated. (6 Marks)
- v) A 1000KVA single transformer has 0.02 p.u resistance, 0.10 p.u reactance and open circuit secondary voltage of 810V. It is connected in parallel with a 250KVA single phase transformer having 0.03 p.u resistance, 0.08 p.u reactance and an open circuit secondary voltage of 415V. Find:
 - (a) Circulating current at no load
 - (b) Currents delivered by the secondary of each transformer to a load of 750KVA at 0.8 lagging pf.
 - (c) KVA and p.f of each transformer (6 Marks)

vi) Determine the no-load parameters of a three-phase oil transformer. The main data of the transformer are:

$P=6000\text{KVA}$, $V_1/V_2 = 40000/7000\text{V}$, $I_1/I_2 = 100/500\text{A}$, $P_o = 20\text{KW}$, $I_o = 5\%$, winding connection $Y/\Delta - 11$

(6 Marks)

Question Two (20 marks)

i) What is the principle of operation of a DC Generator?

(4 Marks)

ii) Explain the three tests performed on transformers.

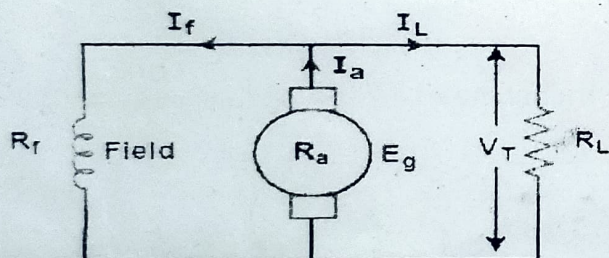
(6 Marks)

iii) A 250V dc motor running at 10 rev/s has an armature resistance of $0.5\ \Omega$. The armature current is 40 A. Determine the torque developed.

(4 Marks)

iv) From the circuit diagram of a dc generator derive the voltage equation.

(6 Marks)



Question Three (20 marks)

i) What is the principle of operation of an Induction Motor?

(4 Marks)

ii) A three-phase induction motor is wound for 4 poles and is supplied from a 50Hz system. Calculate:

a) the synchronous speed

b) the speed of the rotor when the slip is 4%

c) the rotor frequency when the speed of the rotor is 600 r/min

(6 Marks)

iii) A 6 pole, lap wound d.c. motor has 540 conductors with armature resistance of $0.8\ \Omega$. Its speed is found to be 1500 r.p.m. when it is made to run light. The flux per pole is 30mWb. It is connected to a power supply of 250 V. Find the:

a) Induced e.m.f. b) Armature current c) mechanical power d) Armature torque (4 Marks)

iv) A separately-excited generator develops a no-load e.m.f. of 150V at an armature speed of 20 rev/s and a flux per pole of 0.10Wb.

Determine the generated e.m.f. when:

- (a) the speed increases to 25 rev/s and the pole flux remains unchanged
- (b) the speed remains at 20 rev/s and the pole flux is decreased to 0.08Wb.
- (c) the speed increases to 24 rev/s and the pole flux is decreased to 0.07Wb. (6 Marks)

Question Four (20 marks)

i) Differentiate between the three methods of controlling dc motor speed: Field control method, Armature control method and Ward Leonard. (6 Marks)

ii) A 200V d.c. motor develops a shaft torque of 15 Nm at 1200 rev/min. If the efficiency is 80%. Determine the current supplied to the motor. (4 Marks)

iii) A 250V, two pole, 60Hz Y-connected wound rotor induction motor is rated at 20 hp. Its equivalent circuit components are:

$R_1 = 0.3\Omega$	$R_2 = 0.2\Omega$	$X_M = 15\Omega$
$X_1 = 0.52\Omega$	$X_2 = 0.52\Omega$	slip = 5%
$P_{\text{mech}} = 250\text{W}$	$P_{\text{misc}} \approx 0$	$P_{\text{core}} = 150\text{W}$

Calculate:

- (a) The line current I_L
- (b) The stator copper losses P_{SCL}
- (c) The air-gap power P_{AG}
- (d) The power converted from electrical to mechanical P_{conv}
- (e) The induced torque τ_{ind}
- (f) The load torque τ_{load}
- (g) The overall machine efficiency
- (h) The motor speed in revolutions per minute and radians per second (10 Marks)

Question Five (20 marks)

- i) Differentiate between the three types of connections of a three-phase transformer. (6 Marks)
- ii) A 4-pole generator has a lap-wound armature with 50 slots with 16 conductors per slot. The useful flux per pole is 30mWb. Determine the speed at which the machine must be driven to generate an e.m.f. of 240V. (6 Marks)
- iii) A 3-phase step down transformer is connected to 35KV and takes a current of 92.5A. Calculate the secondary line voltage, line current and output for the following connections:
- a) Δ/Δ b) Y/Y c) Δ/Y d) Y/Δ (8 Marks)

$$k=12$$



KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE
(MECHANICAL ENGINEERING)

EMM 216: WORKSHOP PROCESSES AND PRACTICE III

1ST SEMESTER 2022/2023

TIME: 2 Hours

INSTRUCTIONS:

1. Question one is compulsory and carries (20) Twenty marks.
2. Answer any other Two (2) questions from the remaining four.
3. The four questions carry (10) Ten marks each.

Question One

- (a) What is milling and how do you specify a milling machine? (5 marks)
- (b) Give a classification of milling machines. (5 marks)
- (c) List down the types of grinding machines you know. (5 marks)
- (d) What are the advantages of welding over other joints. (5 marks)

Question Two

- (a) A 50mm ^{diameter} ~~delimeter~~ plain milling cutters having 8 teeth is use to face mill a stock of brass 200mm long and 30mm wide. The spindle speed is 1500 rpm and the feed is 0.125 mm/tooth/rev. Find cutting ^{time} ~~line~~. (7 marks)
- (b) What are the common methods of making gears? (3 marks)

Question Three

- (a) Give a list of at least five equipment used for oxyacetylene welding. (5 marks)
- (b) Give a classification of grinding machines (5 marks)

Question Four

- (a) What are the steps to be observed when executing welding? (6 marks)
- (b) Name the various milling attachments (4 marks)

Question Five

- (a) How can one differentiate between up-cut milling and down-cut milling? (6 marks)**
- (b) What is Gas welding process? (4 marks)**



KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE
(MECHANICAL ENGINEERING)
EMM 217: PRINCIPLES OF ENGINEERING DESIGN

1ST SEMESTER, 2022/2023

TIME: 2 Hours

INSTRUCTIONS

1. *This paper contains FIVE questions.*
2. *Question ONE is compulsory and carries 30 marks.*
3. *Four remaining questions carry 20 marks each.*
4. *Attempt question ONE and any other TWO.*

QUESTION 1 (30 MARKS)

- (a) Discuss TWO reasons why the knowledge of materials science is important to mechanical design engineers (2 Marks)
- (b) A tensile-testing apparatus is to be constructed that must withstand a maximum load of 220,000 N. The design calls for two cylindrical support posts, each of which is to support half of the maximum load. Furthermore, plain-carbon (1045) steel ground and polished shafting rounds are to be used; the minimum yield and tensile strengths of this alloy are 310 MPa and 565 MPa respectively. Specify a suitable diameter for these support posts if you are to use a factor of safety of 5 (4 Marks)
- (c) Imagine that you have been hired as a design engineer by a new company and you have been tasked to design and Ugali cooker.
- (i) Explain FOUR questions that will guide your design problem definition (4 Marks)
 - (ii) Discuss FIVE levels of analysis that you would conduct on potential solutions to this design problem (5 Marks)
 - (iii) Using examples, explain how you would design for safety (3 Marks)
 - (iv) Once you build a prototype for the Ugali cooker, you will need to develop a testing protocol to guide its performance evaluation. Specify TEN questions that may form part of your testing protocol (5 Marks)

- (v) Suppose your prototype fails the testing protocol, discuss **TWO** ways that you may opt to proceed with your design (2 Marks)

(d) Differentiate between the following:

- (i) Standards and codes (2 Marks)
(ii) Developmental, adaptive and new design (3 Marks)

QUESTION TWO (20 MARKS)

- (a) Discuss how the **THREE** ergonomic principles learnt in class can be applied to the design of an intensive care unit (ICU) hospital bed (6 Marks)
- (b) Provide **FOUR** reasons why you think standards and codes are important in mechanical engineering design (4 Marks)
- (c) Why is it important to include tolerance to your designs before sending them out for manufacturing? (2 Marks)
- (d) A material has a yield strength, $S_y = 500 \text{ MPa}$ and an ultimate tensile of $S_{ut} = 750 \text{ MPa}$. For $\sigma_1 = 300 \text{ MPa}$, $\sigma_2 = 200 \text{ MPa}$ and $\tau_{max} = 90 \text{ MPa}$, estimate the factor of safety using:
- (i) The maximum shear stress theory (1 Marks)
(ii) The distortion theory (1 Marks)
(iii) The maximum normal stress theory (1 Marks)
- (e) Discuss **FIVE** manufacturing costs that you have to take into account when determining the cost of a design product (5 Marks)

QUESTION THREE (20 MARKS)

- (a) Assume that you are hired as a design engineer for a company that plans to make motorised fishing boats. The company is considering using one of the materials listed in Table Q3(a) to design the hull (body) of the boat. The right candidate should have a minimum yield strength of 250 MPa and a minimum Brinell Hardness Value (BHV) of 99.
- (b)

Table Q3(a)

Material	Yield strength (MPa)	Tensile strength (MPa)	Hardness (BHN)	Cost per kilogram (Kes)
Mild steel	370	440	130	70

Aluminum 6061	276	310	95	400
316L Stainless steel	332	673	230	422

- (i) Using the material selection criteria, explain how you would select the best candidate for use. (10 Marks)
- (ii) Discuss **FOUR** steps that you would take to avoid product liability (4 Marks)
- (iii) Discuss **FOUR** design constraints that you may encounter in this design (6 Marks)

QUESTION FOUR (20 MARKS)

- (a) Engineering design is a highly iterative process. Explain (2 Marks)
- (b) The manufacturability of an engineering product is one of the key design factors that have to be considered early in the design cycle. Discuss **FOUR** practical steps that you can take to ease the manufacturability of an engineering product (8 Marks)
- (c) Noise is a nuisance in design and ought to be eliminated as much as possible. Suppose a part of your design product is making unnecessary noise, what **FOUR** things can you do to resolve the issue? (4 Marks)
 - (i) What is a machine element? (1 Mark)
 - (ii) Differentiate between a clutch and a break (2 Marks)
 - (iii) Explain **THREE** factors that you would consider when selecting a bearing for a given task (3 Marks)

QUESTION FIVE (20 MARKS)

- (a) Discuss **SIX** factors that you would consider when determining the optimum factor of safety for an engineering component or product. (6 Marks)
- (b) Discuss how the information about materials used to make engineering design products is generated and stored. (3 Marks)
- (c) Identify any **THREE** of the biggest issues facing humanity currently that may require an engineering solution and propose possible solutions for each (6 Marks)
- (d) Explain **FIVE** differences between a model and a prototype (5 Marks)



KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE
(MECHANICAL ENGINEERING)
EMM 200: COMPUTER AIDED ENGINEERING DRAWING

1ST SEMESTER 2022/2023

TIME: 2 HOURS

INSTRUCTIONS:

- (i) This paper consists of **FIVE** questions
- (ii) Question **ONE** is compulsory and contains **30** marks
- (iii) Attempt any **TWO** of the remaining four questions. Each question is **20** marks

Question One (30 marks)

- (a) As a young engineer involved in product design, describe any **FOUR** benefits of using CAD to actualize a product of your choice. (4 marks)
- (b) Explain, with the help of diagrams, the functions of the following tools as used in CAD.
 - (i) Orthomode
 - (ii) Trim
 - (iii) Rectangular and polar array (3 marks)
- (c) Describe the **three** Boolean operations commonly used for 3D editing. (3 marks)
- (d) Figure 1 shows the isometric projection of a machine bracket. Using AutoCAD:
 - (i) Redraw the isometric projection. (7 marks)
 - (ii) Draw **three** orthographic views (Front, Top and Right) and show all the principle dimensions. (11 marks)
 - (iii) Use layers to group lines according to functions. (2 marks)

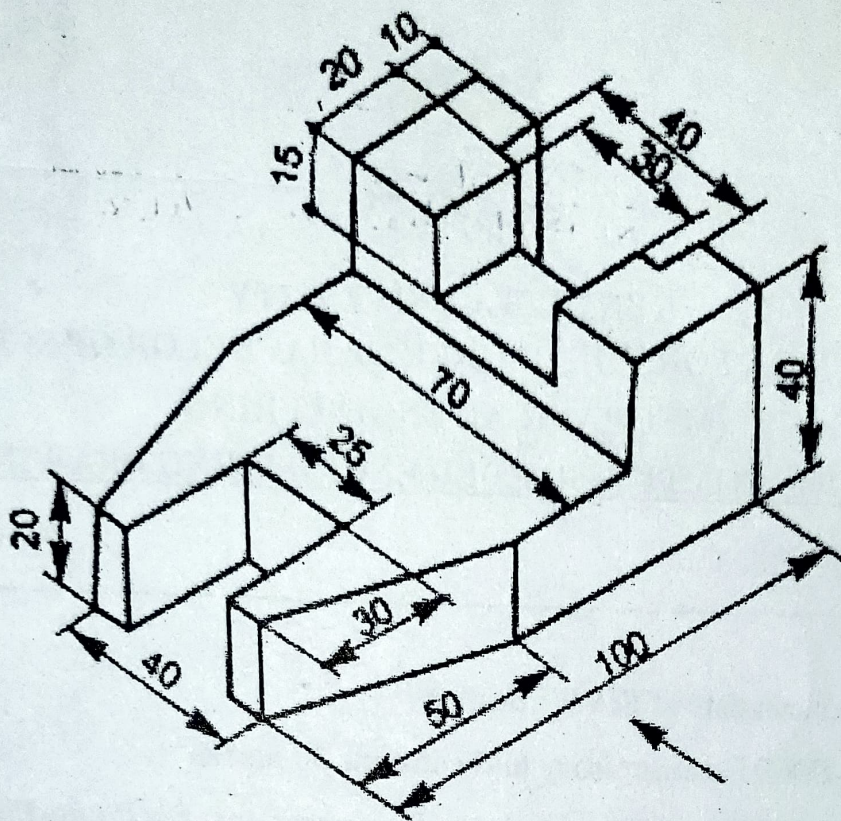


Fig.1

Question two (20 marks)

Figure 2 shows the isometric projection of a machine bracket. Using AutoCAD;

- Produce the isometric view as shown (17 marks)
- ~~Include a suitable title block~~ Using text command input your credentials in the workspace (3 marks)

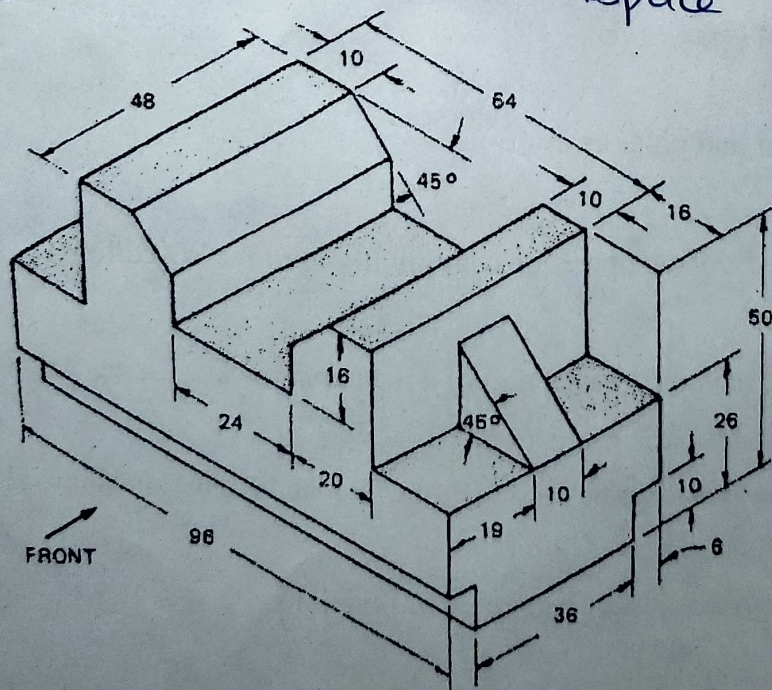


Fig.2

Question three (20 marks)

Figure 3 shows a 3D image, using AUTOCAD:

- Redraw the 3D projection using any suitable methods (17 marks)
- ~~Include a suitable title block~~ Using the text command (3 marks)
input your credentials in the workspace

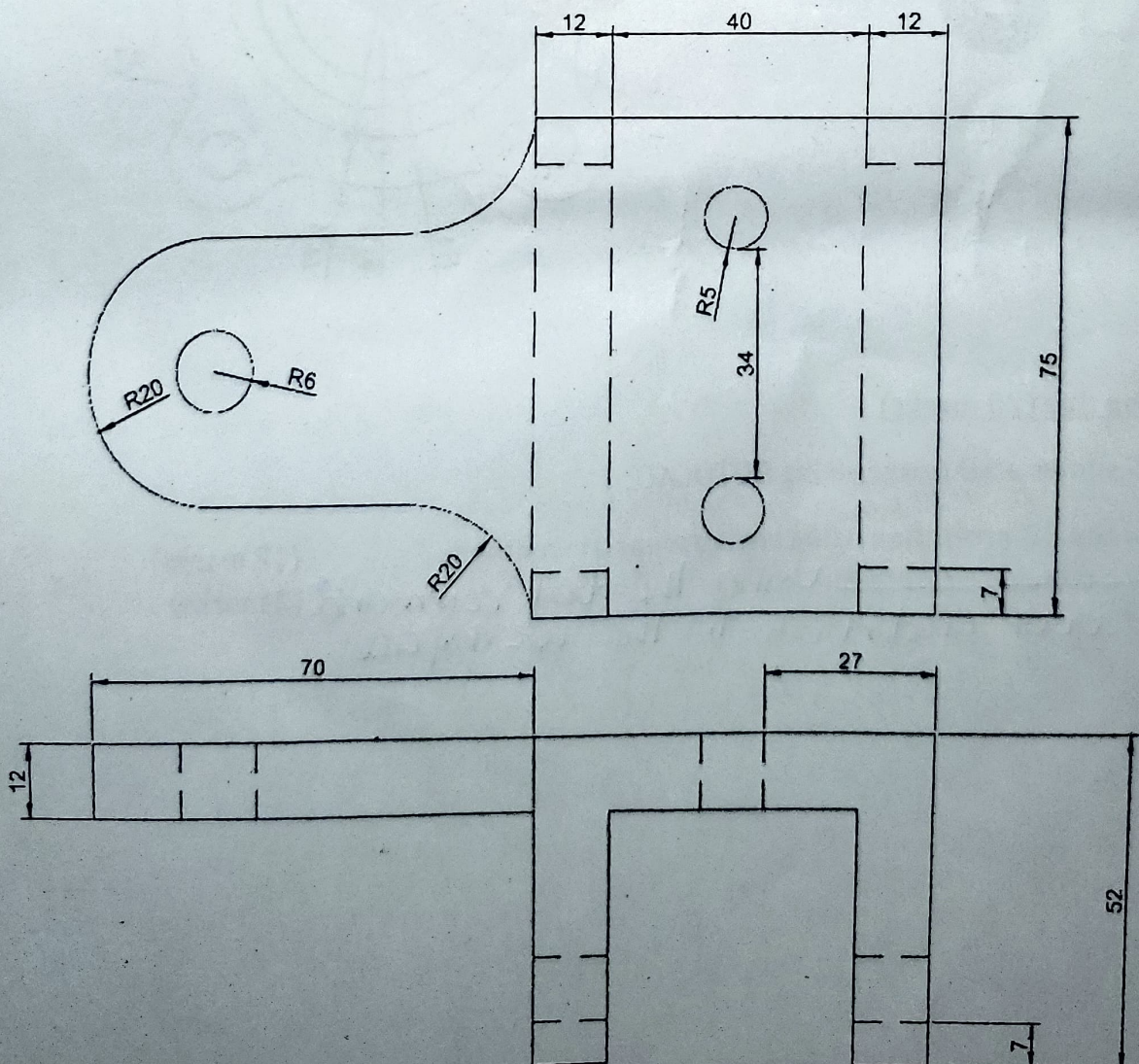
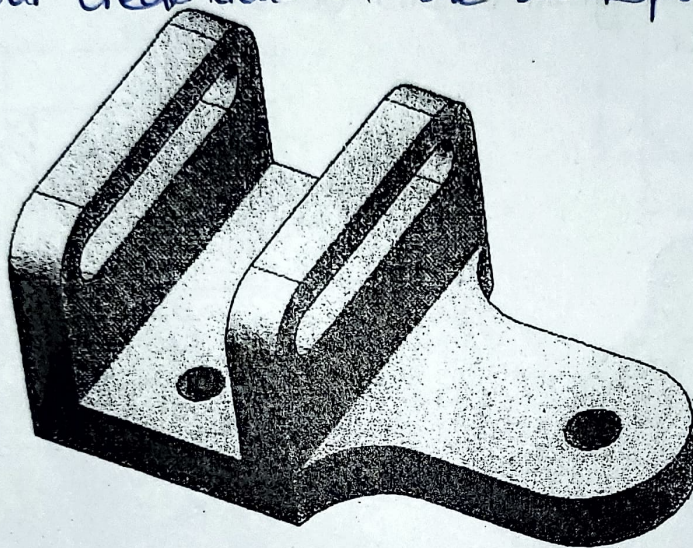


Fig.3

Question four (20 marks)

Figure 4 shows a 3D image, using AUTOCAD:

- Redraw the 3D projection

(17 marks)

- ~~Include a suitable title block~~ Using the text command input your credentials in the workspace

(3 marks)

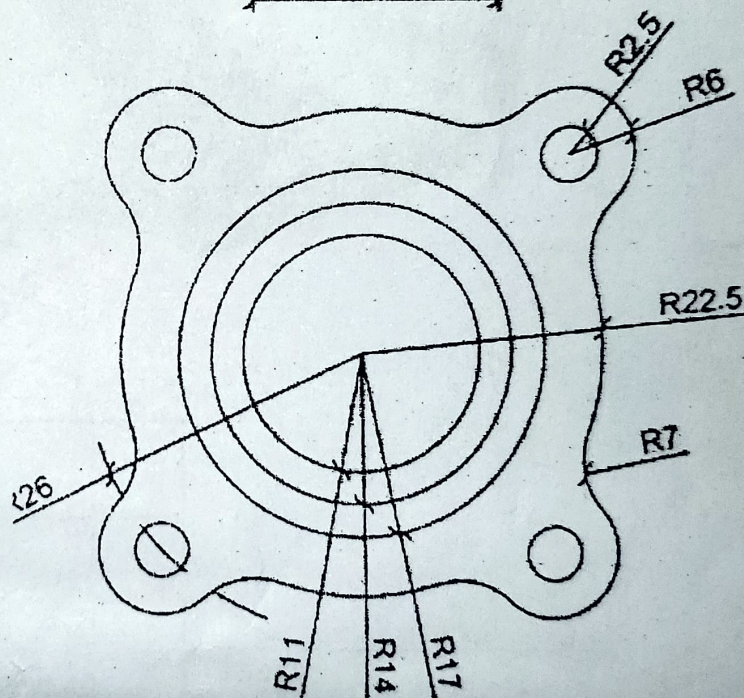
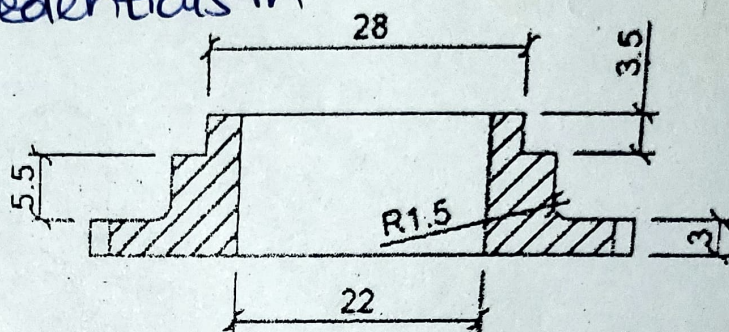
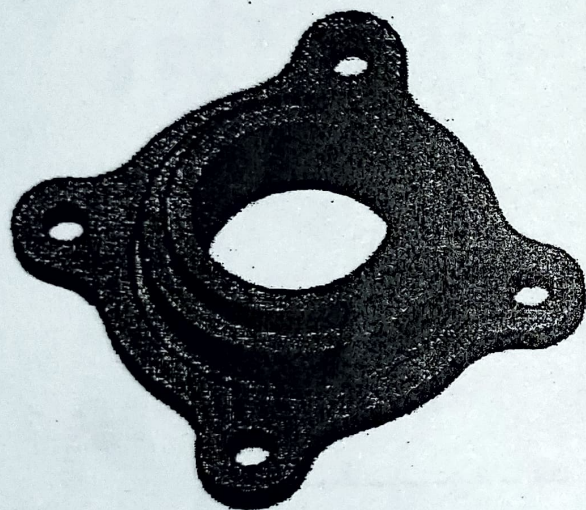


Fig.4

Question five (20 marks)

Figure 5 shows a 3D image, using AUTOCAD:

- Redraw the 3D projection using any appropriate methods

(17 marks)

- ~~Include a suitable title block~~ Using the text command input your credentials in the workspace

(3 marks)

